



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECP3623 – DATABASE PROGRAMMING

SECTION 1

MONGODB EXERCISE (INDIVIDUAL)

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SECP3623 – DATABASE PROGRAMMING
SEM I – 2025/2026
MongoDB(shopdb.orders)

Instruction:

Insert the following data into the shopdb database and the orders collection.

use shopdb

```
db.orders.insertMany([
  {
    order_id: "ORD101",
    customer: "Ali Ahmad",
    product: "Laptop",
    quantity: 1,
    price: 3500,
    status: "Completed",
    order_date: new Date("2025-01-05")
  },
  {
    order_id: "ORD102",
    customer: "Siti Aminah",
    product: "Smartphone",
    quantity: 2,
    price: 2800,
    status: "Pending",
    order_date: new Date("2025-01-06")
  },
  {
    order_id: "ORD103",
    customer: "John Lee",
    product: "Tablet",
    quantity: 1,
    price: 1800,
    status: "Completed",
    order_date: new Date("2025-01-07")
  },
  {
    order_id: "ORD104",
    customer: "Nur Aisyah",
    product: "Laptop",
    quantity: 1,
```

```

    price: 3700,
    status: "Cancelled",
    order_date: new Date("2025-01-08")
  },
  {
    order_id: "ORD105",
    customer: "Ahmad Zaki",
    product: "Headphones",
    quantity: 3,
    price: 450,
    status: "Completed",
    order_date: new Date("2025-01-09")
  },
  {
    order_id: "ORD106",
    customer: "Farah Izzati",
    product: "Smartwatch",
    quantity: 2,
    price: 1200,
    status: "Pending",
    order_date: new Date("2025-01-10")
  }
]

```

- a) Write a MongoDB command to display all orders with status "Completed".
- b) Write a MongoDB command to display all orders with price greater than RM2000.
- c) Write a MongoDB command to display all orders sorted by price in descending order.
- d) Write a MongoDB command to display all "Pending" orders sorted by order_date (latest first).
- e) Using aggregation, calculate the total sales amount (price) for each product.
- f) Using aggregation, calculate the total sales amount for orders with status "Completed".
- g) Write a MongoDB command to update the status of order ORD102 from "Pending" to "Completed".
- h) Write a MongoDB command to increase the quantity of order ORD105 by 1.
- i) Write a MongoDB command to delete the order with order_id "ORD104".
- j) Write a MongoDB command to delete all orders with status "Cancelled".

a) Write a MongoDB command to display all orders with status "Completed".

db.orders.find({status:"Completed"})

```
> db.orders.find({status:"Completed"})
< {
  _id: ObjectId('695f180e928f17e611c64c89'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8b'),
  order_id: 'ORD103',
  customer: 'John Lee',
  product: 'Tablet',
  quantity: 1,
  price: 1800,
  status: 'Completed',
  order_date: 2025-01-07T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8d'),
  order_id: 'ORD105',
  customer: 'Ahmad Zaki',
  product: 'Headphones',
  quantity: 3,
  price: 450,
  status: 'Completed',
  order_date: 2025-01-09T00:00:00.000Z
}
```

b) Write a MongoDB command to display all orders with price greater than RM2000.

`db.orders.find({price:{gt:2000}})`

```
> db.orders.find({price:{gt:2000}})
< {
  _id: ObjectId('695f180e928f17e611c64c89'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8a'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8c'),
  order_id: 'ORD104',
  customer: 'Nur Aisyah',
  product: 'Laptop',
  quantity: 1,
  price: 3700,
  status: 'Cancelled',
  order_date: 2025-01-08T00:00:00.000Z
}
```

c) Write a MongoDB command to display all orders sorted by price in descending order.

`db.orders.find().sort({price:-1})`

```
> db.orders.find().sort({price:-1})
< {
  _id: ObjectId('695f180e928f17e611c64c8c'),
  order_id: 'ORD104',
  customer: 'Nur Aisyah',
  product: 'Laptop',
  quantity: 1,
  price: 3700,
  status: 'Cancelled',
  order_date: 2025-01-08T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c89'),
  order_id: 'ORD101',
  customer: 'Ali Ahmad',
  product: 'Laptop',
  quantity: 1,
  price: 3500,
  status: 'Completed',
  order_date: 2025-01-05T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8a'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8b'),
  order_id: 'ORD103',
  customer: 'John Lee',
  product: 'Tablet',
  quantity: 1,
  price: 1800,
  status: 'Completed',
  order_date: 2025-01-07T00:00:00.000Z
}
```

```
{
  _id: ObjectId('695f180e928f17e611c64c8e'),
  order_id: 'ORD106',
  customer: 'Farah Izzati',
  product: 'Smartwatch',
  quantity: 2,
  price: 1200,
  status: 'Pending',
  order_date: 2025-01-10T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8d'),
  order_id: 'ORD105',
  customer: 'Ahmad Zaki',
  product: 'Headphones',
  quantity: 3,
  price: 450,
  status: 'Completed',
  order_date: 2025-01-09T00:00:00.000Z
}
```


d) Write a MongoDB command to display all "Pending" orders sorted by order_date (latest first).

db.orders.find({status: "pending"}).sort({order_date: -1})

```
> db.orders.find({status:"Pending"}).sort({order_date:-1})
< {
  _id: ObjectId('695f180e928f17e611c64c8e'),
  order_id: 'ORD106',
  customer: 'Farah Izzati',
  product: 'Smartwatch',
  quantity: 2,
  price: 1200,
  status: 'Pending',
  order_date: 2025-01-10T00:00:00.000Z
}
{
  _id: ObjectId('695f180e928f17e611c64c8a'),
  order_id: 'ORD102',
  customer: 'Siti Aminah',
  product: 'Smartphone',
  quantity: 2,
  price: 2800,
  status: 'Pending',
  order_date: 2025-01-06T00:00:00.000Z
}
```

e) Using aggregation, calculate the total sales amount (price) for each product.

```
db.orders.aggregate([
  { $group: { _id: "$product", totalSales: { $sum: "$price" } } }
])
```

```
> db.orders.aggregate([
  { $group: { _id: "$product", totalSales: { $sum: "$price" } } }
])
< {
  _id: 'Headphones',
  totalSales: 450
}
{
  _id: 'Smartphone',
  totalSales: 2800
}
{
  _id: 'Laptop',
  totalSales: 7200
}
{
  _id: 'Tablet',
  totalSales: 1800
}
{
  _id: 'Smartwatch',
  totalSales: 1200
}
```


f) Using aggregation, calculate the total sales amount for orders with status "Completed".

```
db.orders.aggregate([
  { $match: { status: "Completed" } },
  { $group: { _id: null, totalCompletedSales: { $sum: "$price" } } }
])
```

```
> db.orders.aggregate([
  { $match: { status: "Completed" } },
  { $group: { _id: null, totalCompletedSales: { $sum: "$price" } } }
])
< {
  _id: null,
  totalCompletedSales: 5750
}
```

g) Write a MongoDB command to update the status of order ORD102 from "Pending" to "Completed".

```
db.orders.updateOne(  
  {order_id: "ORD102"},  
  { $set: {status: "Completed"} }  
)
```

```
> db.orders.updateOne(  
  {order_id: "ORD102"},  
  { $set: {status: "Completed"} }  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

h) Write a MongoDB command to increase the quantity of order ORD105 by 1.

```
db.orders.updateOne(  
  {order_id: "ORD102"},  
  {$inc: {quantity: 1}}  
)
```

```
> db.orders.updateOne(  
  {order_id: "ORD105"},  
  {$inc: {quantity: 1}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

i) Write a MongoDB command to delete the order with order_id "ORD104".

db.orders.deleteOne({order_id: "ORD104"})

```
> db.orders.deleteOne({order_id:"ORD104"})
< {
  acknowledged: true,
  deletedCount: 1
}
```

j) Write a MongoDB command to delete all orders with status "Cancelled".

db.orders.deleteMany({status:"Cancelled"})

```
> db.orders.deleteMany({status:"Cancelled"})
< {
  acknowledged: true,
  deletedCount: 0
}
```